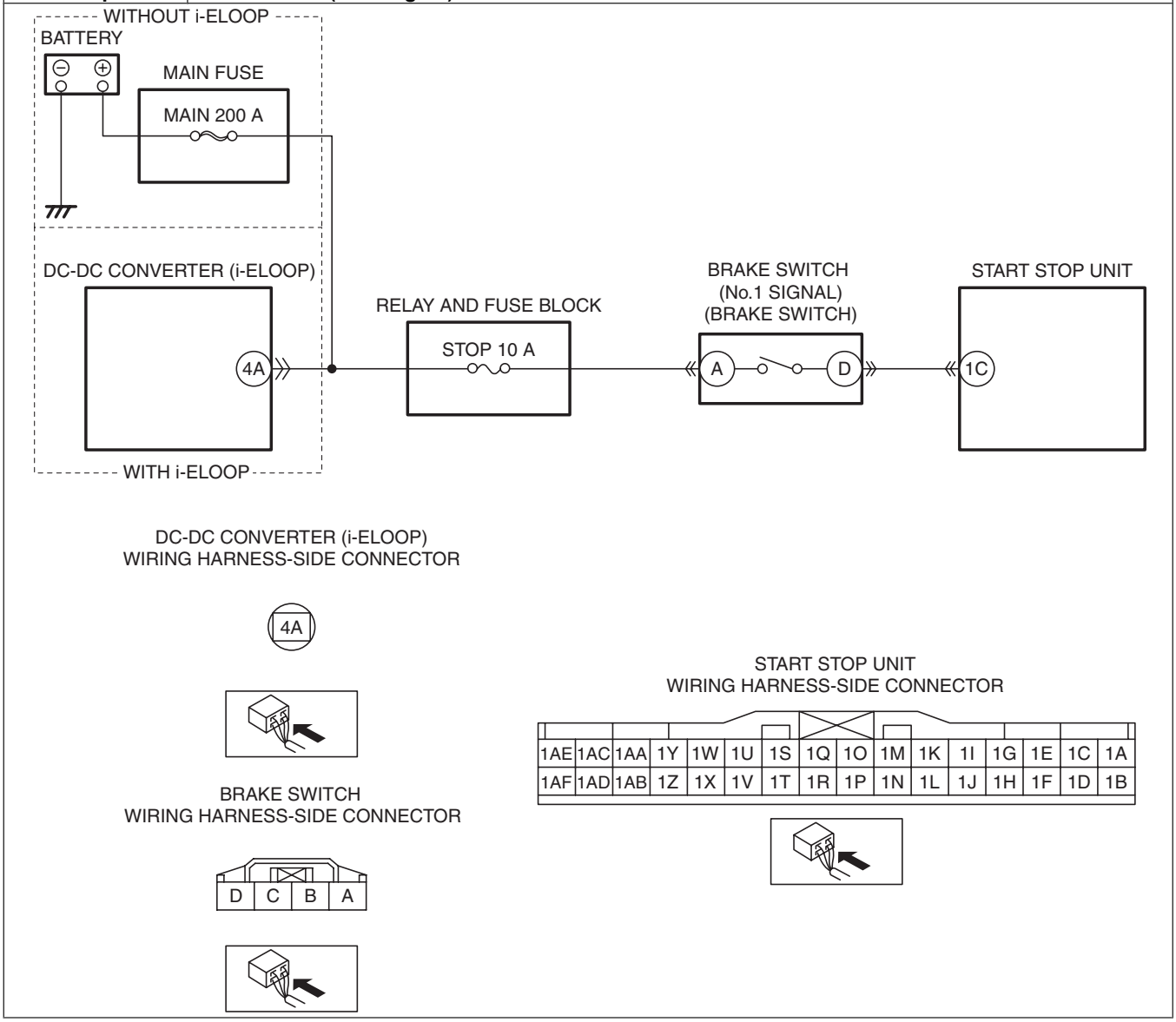
DTC C0040:23 [START STOP UNIT]

id0902p6021100

Description	Brake switch (No.1 signal) circuit malfunction
Detection condition	• The start stop unit detected the following condition **5 times** continuously with the ignition switched ON (engine off or on). — The brake switch (No.1 signal) is off for a continuous **3 s or more** even though the brake switch (No. 2 signal) changes.
Fail-safe	• Determined by brake switch (No.2 signal) with the brake pedal depressed.
Possible cause	• DTCs are stored in the DC-DC converter (i-ELOOP). — DC-DC converter (i-ELOOP) connector or terminal malfunction — DC-DC converter (i-ELOOP) malfunction • Brake switch connector or terminal malfunction • Brake switch (No.1 signal) malfunction • Brake switch (No.1 signal) power supply circuit malfunction — Without i-ELOOP: • Short to ground in wiring harness between MAIN 200 A fuse and brake switch terminal A • STOP 10 A fuse malfunction • MAIN 200 A fuse malfunction • Open circuit in wiring harness between battery positive terminal and brake switch terminal A — With i-ELOOP: • Short to ground in wiring harness between STOP 10 A fuse and brake switch terminal A • STOP 10 A fuse malfunction • Open circuit in wiring harness DC-DC converter (i-ELOOP) terminal 4A and brake switch terminal A • Start stop unit connector or terminal malfunction • Short to ground in wiring harness between brake switch terminal D and start stop unit terminal 1C • Open circuit in wiring harness between brake switch terminal D and start stop unit terminal 1C • Start stop unit malfunction

Description	Brake switch (No.1 signal) circuit malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY DC-DC CONVERTER (i-ELOOP) DTCs** **Note** • For vehicles without i-ELOOP, go to the next step. • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].)
		No	Go to the next step.

Step	Inspection		Action
2	**INSPECT BRAKE SWITCH CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the brake switch connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 9.
3	**INSPECT BRAKE SWITCH (No.1 SIGNAL)** • Inspect the brake switch (No.1 signal). (See BRAKE SWITCH INSPECTION.) • Is the brake switch (No.1 signal) normal?	Yes	**Without i-ELOOP:** • Go to the next step. **With i-ELOOP:** • Go to Step 5.
		No	Replace the brake switch, then go to Step 9. (See BRAKE PEDAL REMOVAL/INSTALLATION [L.H.D.].) (See BRAKE PEDAL REMOVAL/INSTALLATION [R.H.D.].)
4	**INSPECT BRAKE SWITCH (No.1 SIGNAL) POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT OR SHORT TO GROUND** • Verify that the brake switch connector is disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at the brake switch terminal A (wiring harness-side). • Is the voltage **B+**?	Yes	Go to Step 5.
		No	Inspect the STOP 10 A fuse and MAIN 200 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and brake switch terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and brake switch terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.

Step	Inspection		Action
5	**INSPECT BRAKE SWITCH (No.1 SIGNAL) POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT OR SHORT TO GROUND** • Verify that the brake switch connector is disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at the brake switch terminal A (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Disconnect the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Inspect the STOP 10 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and brake switch terminal A. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and brake switch terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Connect the service plug, then go to Step 9. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)
6	**INSPECT START STOP UNIT CONNECTOR CONDITION** • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the start stop unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 9.

Step	Inspection		Action
7	**INSPECT BRAKE SWITCH (No.1 SIGNAL) CIRCUIT FOR SHORT TO GROUND** • Verify that the brake switch and start stop unit connectors are disconnected. • Inspect for continuity between brake switch terminal D (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between brake switch terminal D and start stop unit terminal 1C. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 9.
		No	Go to the next step.
8	**INSPECT BRAKE SWITCH (No.1 SIGNAL) CIRCUIT FOR OPEN CIRCUIT** • Verify that the brake switch and start stop unit connectors are disconnected. • Inspect for continuity between brake switch terminal D (wiring harness-side) and start stop unit terminal 1C (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between brake switch terminal D and start stop unit terminal 1C. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
9	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].) • Perform the work of depressing the brake pedal for **3 s or more** and then releasing it for **5 times or more** with the ignition switched ON (engine off or on). • Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
10	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [START STOP UNIT].)
		No	DTC troubleshooting completed.